

# Azure Load Balancer Deployment

**Title:** Creating and Verifying Internal & External Load Balancers in Microsoft Azure

**Student Name:** Tejasvi Manjit Avhad

**Internship Domain:** Cloud, Infrastructure, and Security

**Tool Used:** Microsoft Azure

## 1.Introduction:

In cloud computing environments, load balancing plays a vital role in distributing network traffic efficiently across multiple servers to ensure application reliability and performance. This project focuses on implementing both an Internal and External Load Balancer using Microsoft Azure. The goal is to simulate real-world traffic distribution by configuring a pair of virtual machines (VMs) behind both public and private load balancers. This not only helps in understanding traffic routing strategies but also provides hands-on experience with Azure networking components such as VNets, subnets, backend pools, probes, and load balancing rules.

## 2.Objective:

To deploy and configure both an Internal and External Load Balancer in Microsoft Azure using two virtual machines (VMs) in a Virtual Network (VNet) to demonstrate basic traffic distribution and load balancing principles.

## 3.Resource Planning:

| Resource Type   | Name                | Region      |
|-----------------|---------------------|-------------|
| Resource Group  | LoadBalancerProject | East US     |
| Virtual Network | ProjectVNet         | South India |
| Subnet          | AppSubnet           | South India |
| VM1             | AppVM1              | South India |
| VM2             | AppVM2              | South India |
| Public IP       | PublicIPExt         | South India |
| External LB     | ExternalLB          | South India |
| Internal LB     | InternalLB          | South India |

## 4.Implementation:

STEP 1: Create Resource Group

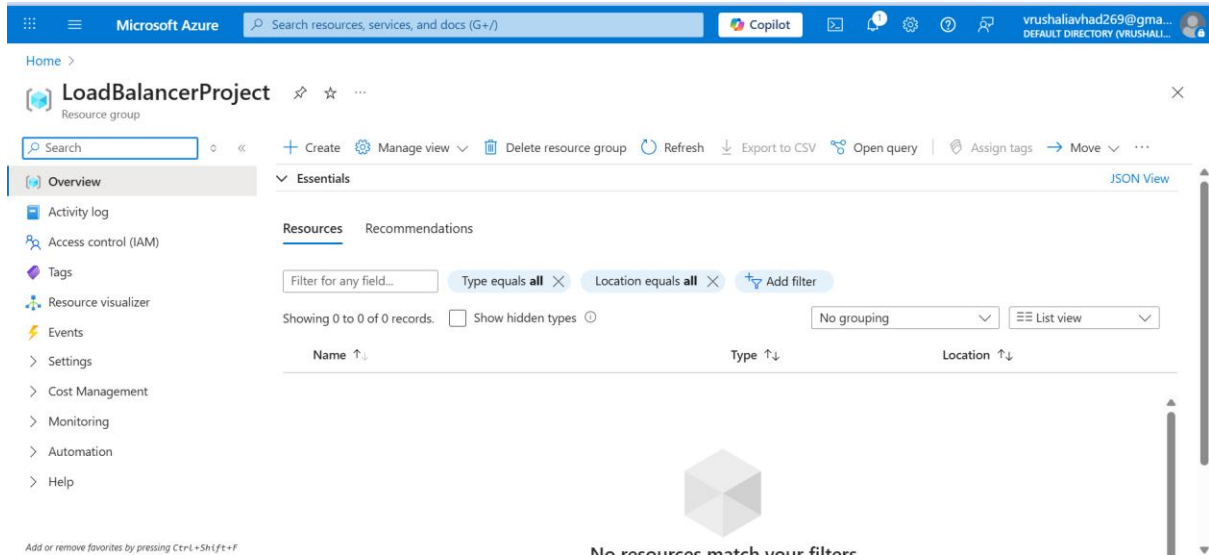
1. Go to: <https://portal.azure.com>

2. Search "Resource Groups" → Click + Create

3. Fill:

- Name: LoadBalancerProject
- Region: East US

4. Click Review + Create → Create



## STEP 2: Create Virtual Network

Search “Virtual Networks” → Click + Create

1. Basics:

- Resource Group: LoadBalancerProject
- Name: ProjectVNet
- Region: South India

2. IP Address tab:

- Leave Address Space: 10.0.0.0/16
- Subnet name: AppSubnet
- Subnet range: 10.0.0.0/24

3. Click Review + Create → Create

The screenshot shows the 'Create virtual network' wizard in the Microsoft Azure portal, specifically the 'Review + create' tab. The breadcrumb navigation at the top indicates the path: Home > Network foundation | Virtual networks >. The page title is 'Create virtual network'. Below the title are tabs for 'Basics', 'Security', 'IP addresses', 'Tags', and 'Review + create'. The 'Basics' section contains the following configuration details:

| Basics         |                     |
|----------------|---------------------|
| Subscription   | Azure for Students  |
| Resource Group | LoadBalancerProject |
| Name           | ProjectVNet         |
| Region         | South India         |

The 'Security' section contains the following configuration details:

| Security                      |          |
|-------------------------------|----------|
| Azure Bastion                 | Disabled |
| Azure Firewall                | Disabled |
| Azure DDoS Network Protection | Disabled |

At the bottom of the wizard, there are buttons for 'Previous', 'Next', and 'Create'. A link 'Download a template for automation' is also present. In the bottom right corner, there is a 'Give feedback' link.

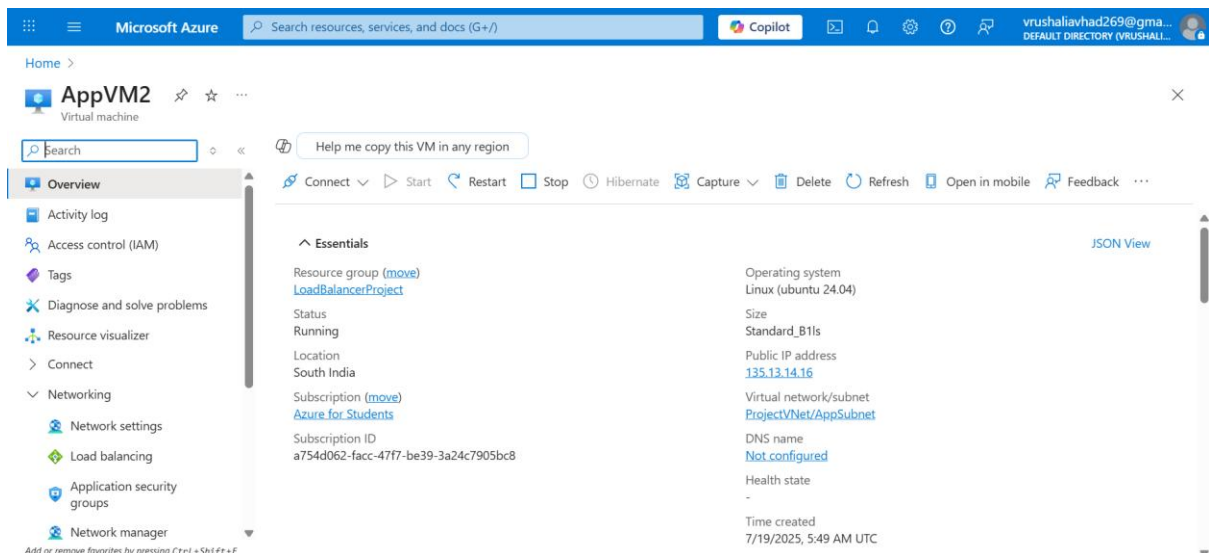
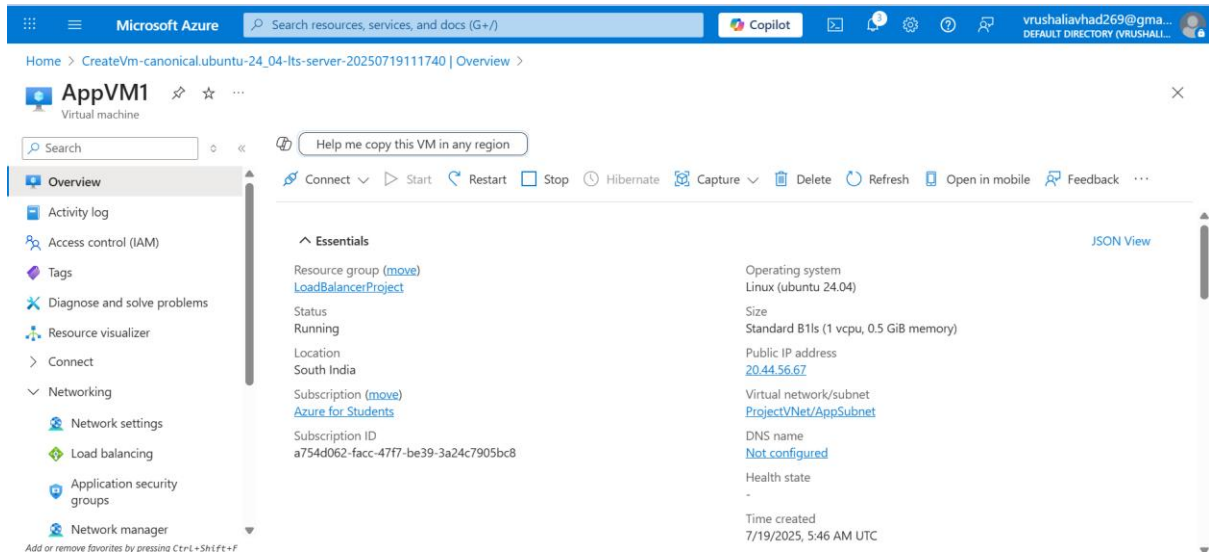
### STEP 3: Create Two Virtual Machines (VMs)

Repeat the steps below for both AppVM1 and AppVM2

For AppVM1:

1. Go to Virtual Machines → Click + Create
2. Basics:
  - Name: AppVM1
  - Region: South India
  - Image: Ubuntu Server 20.04 LTS
  - Size: Standard B1s (cheapest)
  - Username: azureuser
  - Password: Strong password (e.g. MyPassword@123)
3. Inbound Ports: None
4. Networking:
  - VNet: ProjectVNet
  - Subnet: AppSubnet
  - Public IP: None
5. Click Review + Create → Create

Repeat same steps for AppVM2



## STEP 4: Install Nginx Web Server

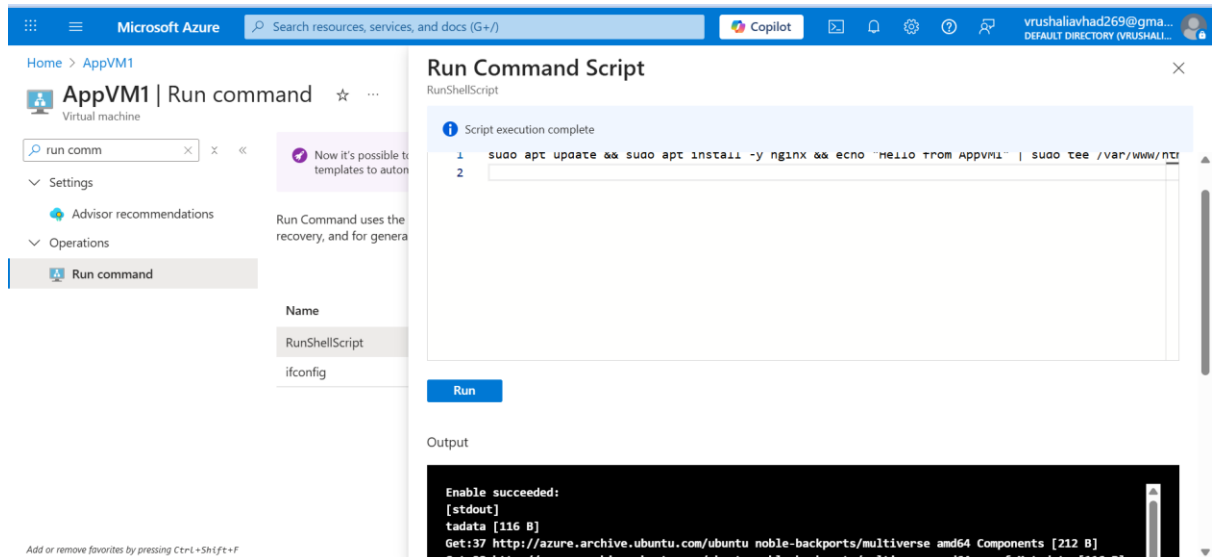
Use Azure's Run Command on each VM.

For AppVM1:

1. Go to Virtual Machines → Open AppVM1
2. Left panel → Click Run Command
3. Choose RunShellScript
4. Paste this:

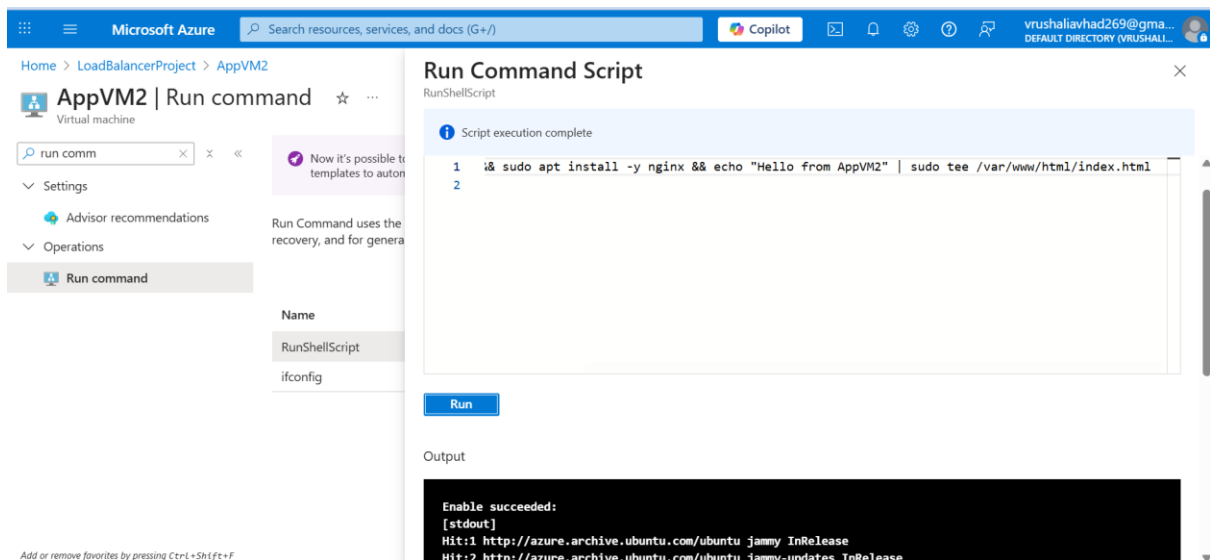
```
sudo apt update && sudo apt install -y nginx && echo "Hello from AppVM1" | sudo tee /var/www/html/index.html
```

5. Click Run



For AppVM2:

```
sudo apt update && sudo apt install -y nginx && echo "Hello from AppVM2" | sudo tee /var/www/html/index.html
```



## STEP 5: Create External Load Balancer

1. Search "Load Balancers" → Click + Create
2. Basics:
  - Name: ExternalLB
  - Region: South India
  - Type: Public
  - SKU: Standard

- Resource Group: LoadBalancerProject
3. Frontend IP:
    - Click + Add frontend IP
    - Name: ExtFrontend
    - Public IP: Create new → Name it PublicIPExt
  4. Click Next: Backend pools

#### Backend Pool (External LB)

1. Click + Add
  - Name: ExtBackendPool
  - VNet: ProjectVNet
2. Under Add targets:
  - Add AppVM1 → NIC → ipconfig1
  - Add AppVM2 → NIC → ipconfig1
3. Click Add

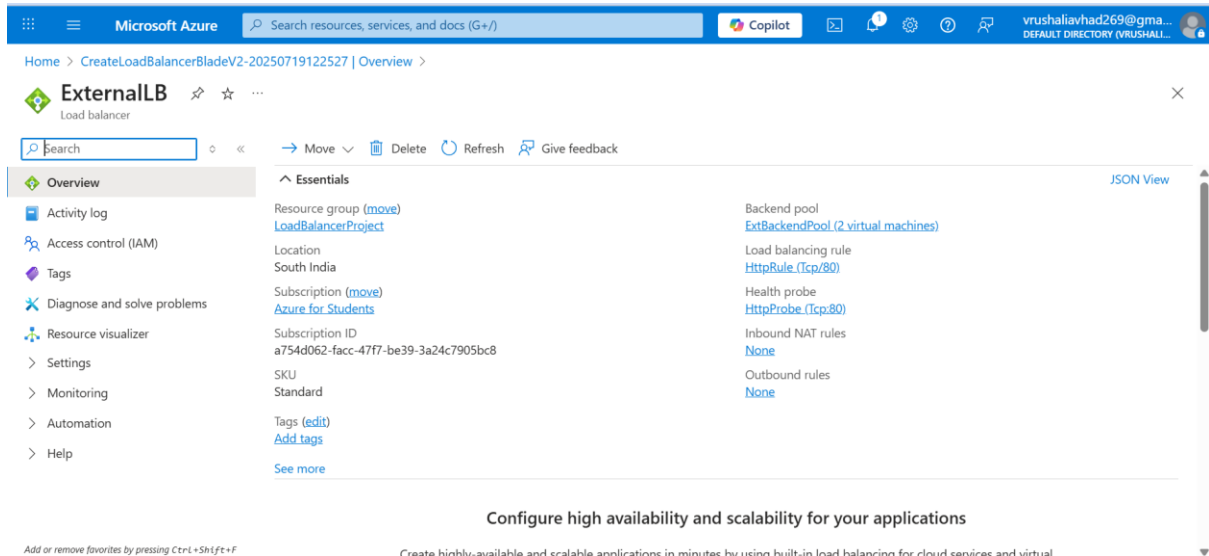
#### Health Probe

1. Click + Add
  - Name: HttpProbe
  - Protocol: HTTP
  - Port: 80
  - Path: /
2. Click Add

#### Load Balancing Rule

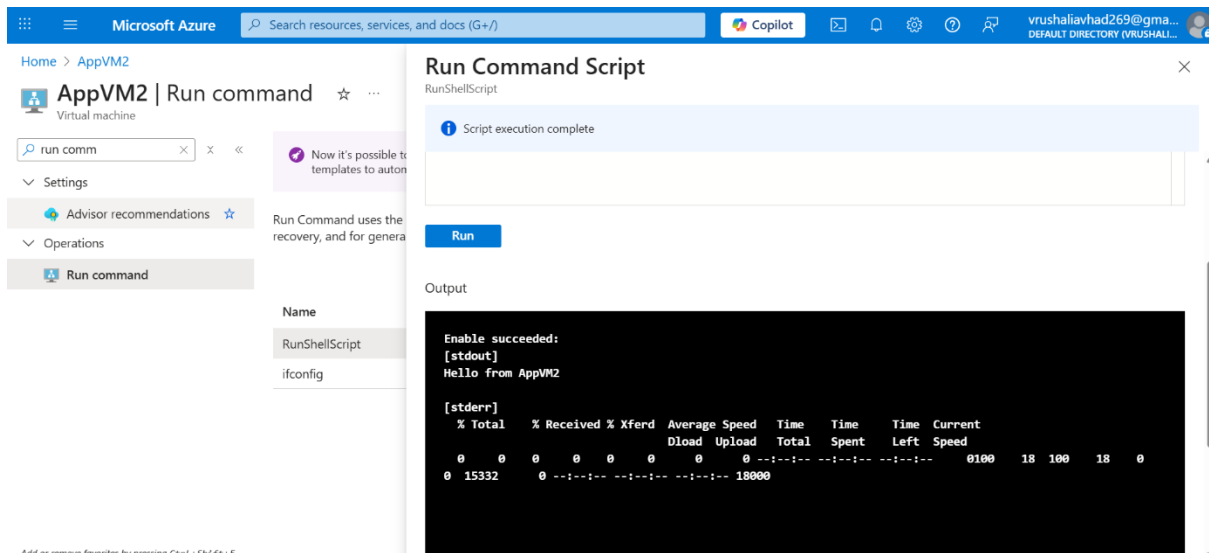
1. Click + Add
  - Name: HttpRule
  - Frontend: ExtFrontend
  - Backend pool: ExtBackendPool
  - Protocol: TCP

- Port: 80 → 80
  - Probe: HttpProbe
2. Click Add
  3. Click Review + Create → Create



## STEP 6: Test External Load Balancer

1. Go to Public IP addresses → Click PublicIPExt
2. Copy IP address → Paste in browser
3. You should see:
  - “Hello from AppVM1” OR
  - “Hello from AppVM2”
4. Refresh multiple times → Response should alternate



## STEP 7: Create Internal Load Balancer

1. Go to Load Balancers → Click + Create
2. Basics:
  - Name: InternalLB
  - Type: Internal
  - SKU: Standard
  - Region: South India
  - Resource Group: LoadBalancerProject
3. Frontend IP:
  - Name: IntFrontend
  - Private IP: Static → use: 10.0.0.10
  - VNet: ProjectVNet
  - Subnet: AppSubnet
4. Click Next: Backend pools

### Backend Pool (Internal LB)

1. Name: IntBackendPool
2. Add AppVM1 and AppVM2 NICs (ipconfig1)
3. Click Next: Health Probe



### Health Probe (Same as before)

- Name: HttpProbe
- Protocol: HTTP
- Port: 80
- Path: /

### Load Balancing Rule

1. Name: HttpRule
2. Frontend IP: IntFrontend
3. Backend pool: IntBackendPool
4. Port: 80 → 80
5. Probe: HttpProbe
6. Click Review + Create → Create

Microsoft Azure Search resources, services, and docs (G+)

Home > Load balancing and content delivery | Load balancers >

### Create load balancer

Validation passed

Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

**Basics**

|                |                     |
|----------------|---------------------|
| Subscription   | Azure for Students  |
| Resource group | LoadBalancerProject |
| Name           | InternalLB          |
| Region         | South India         |
| SKU            | Standard            |
| Tier           | Regional            |
| Type           | Internal            |

Frontend IP configuration

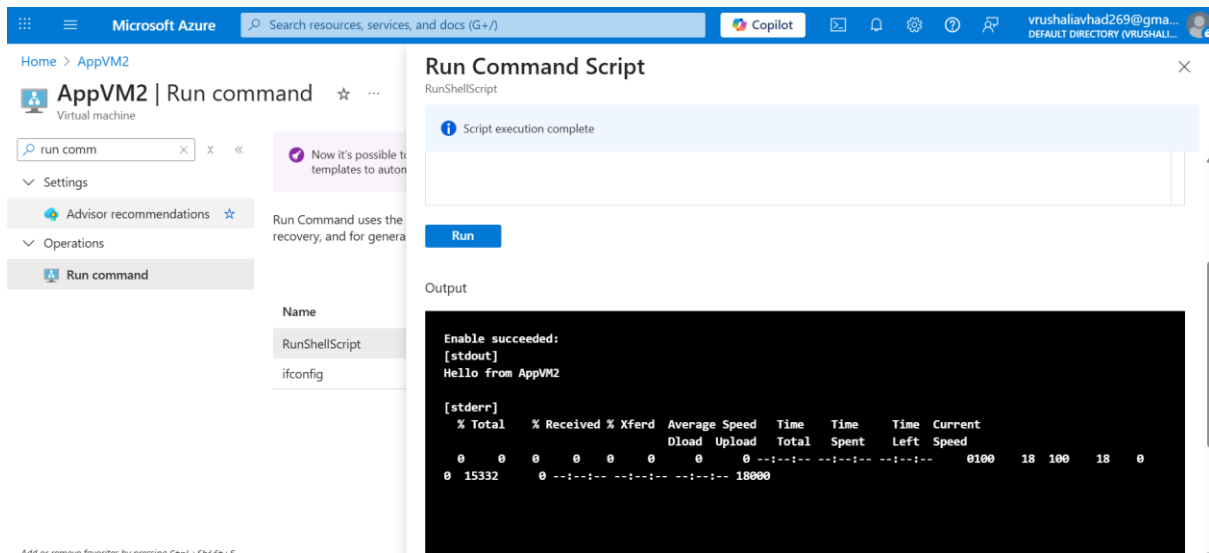
Create < Previous Next > Download a template for automation Give feedback

### STEP 8: Test Internal Load Balancer

From inside AppVM2, use Run Command:

`curl http://10.0.0.10`

- Output should be:
  - “Hello from AppVM1” or “Hello from AppVM2”
- Run again to see alternating responses



## 5.Verification:

- External LB: Public IP opened in browser showed messages from both VMs alternately.
- Internal LB: Used curl command from one VM to private IP. Got expected response from both VMs.

## 6.Conclusion:

- Successfully deployed and tested both Internal and External Load Balancers.
- Demonstrated knowledge of Azure networking, VM setup, backend pools, health probes, and load balancing rules.